

ECA0905-DIGITAL SIGNAL PROCESSING

DATE:24-07-2026

EXPERIMENT 5: Design of High Pass Filter using Windowing Techniques.

5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

PROGRAM

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);

%% --- 2. FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---

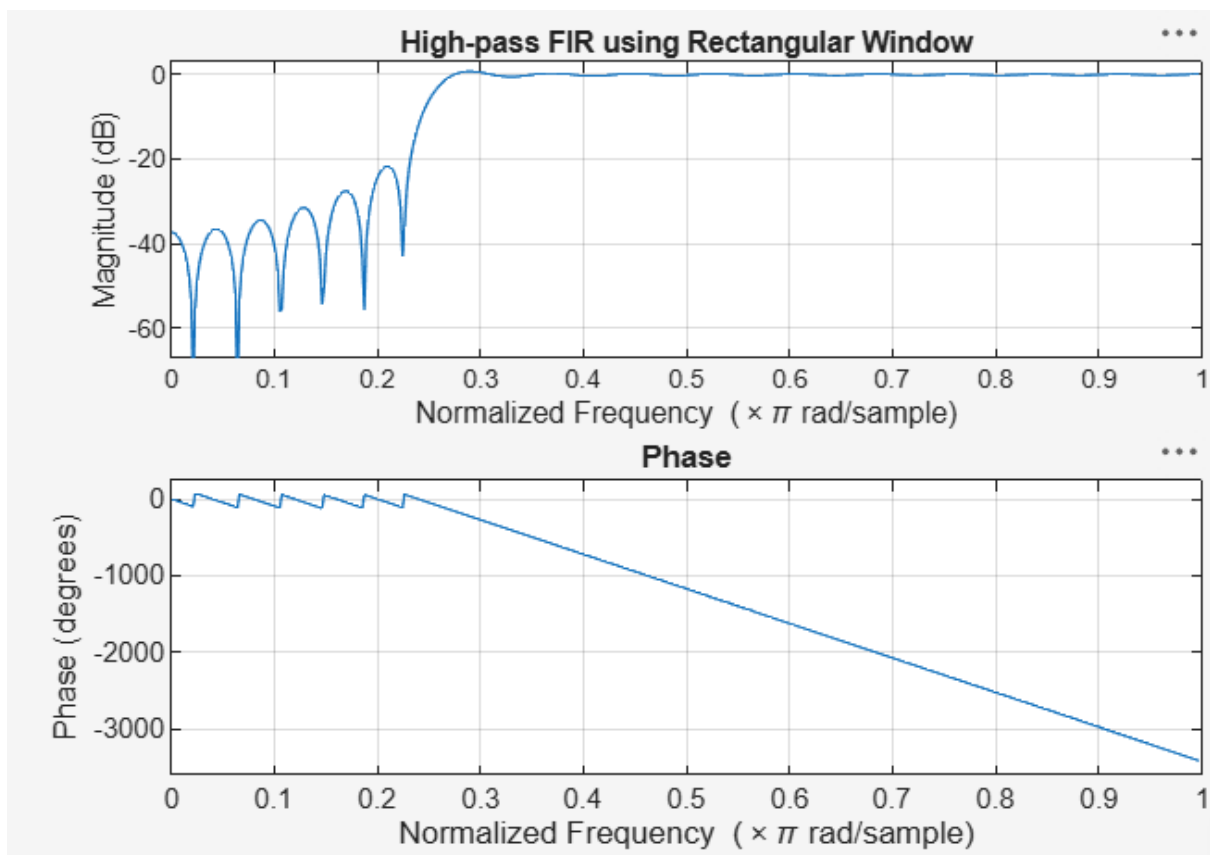
figure;
```

```

freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');

```

OUTPUT



5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

PROGRAM